

SOP-105 — Shell-and-Tube Heat Exchanger Cleaning and Inspection

Plant Site A · Rev 3 · Issued 05/06/2025 · Approved by Plant Engineering

1. Purpose

This procedure defines the method for isolating, opening, cleaning and inspecting the shell-and-tube heat exchangers at Plant Site A. Fouling of the tube bundle is the dominant cause of lost thermal performance and of overheating events on downstream equipment, and periodic cleaning restores the design heat transfer coefficient.

2. Scope

Applies to exchangers HX-201A and HX-201B in the Process Area and HX-202A in the Utilities Area. It covers routine mechanical cleaning of the tube side and shell side, visual inspection, and return to service. Retubing, plugging beyond ten percent of tubes, and pressure test failures are escalated to the static equipment engineer.

3. References

SOP-118 Equipment Isolation and Lock-Out Tag-Out.

Exchanger general arrangement and tube layout drawings for the unit being worked.

TEMA standards guidance section, held in the technical library.

4. Safety Precautions

Both shell side and tube side shall be isolated, drained, depressurised and proven dead before any flange is broken. Hot condensate hazards persist long after isolation; verify metal temperature is below 45 degrees C before starting work.

Channel head and floating head covers are heavy eccentric loads. Use the davit or certified rigging; never support a head on the stud threads.

High pressure water jetting above 250 bar shall only be performed by certified operators inside a barricaded exclusion zone.

5. Tools and Materials

Have available: the davit or certified rigging with current test certificates, hydraulic torque equipment covering the flange stud sizes, the bolting table for the unit, new gaskets of the specified material for every joint to be broken, tube cleaning lances and nozzles matched to the tube bore, a borescope, rated blanking spades for the isolation, and trays and drums for the deposits removed. High pressure jetting units shall carry a current inspection certificate, and operator certification shall be verified before work starts.

Verify the exchanger identity against the work order before isolation. HX-201A and HX-201B sit side by side in the structure, and opening the in-service unit of the pair is a serious incident.

6. Procedure

6.1 Confirm isolation of all four nozzles per SOP-118, drain both sides to the closed drain system, and vent. Verify zero pressure at the vent and drain before unbolting.

6.2 Remove the channel head cover using the davit. Inspect the channel, partition plate and gasket faces for erosion, corrosion and gasket imprint condition, and photograph as found.

6.3 Record the as-found fouling: estimate the percentage of tube ends blocked and the deposit type, whether soft biological film, hard scale, or process polymer. This record drives the cleaning interval review.

6.4 Clean the tube side by high pressure water jetting each tube with the correct nozzle size, working the full tube length in both directions until the lance passes freely and returns clear water.

6.5 Where shell side cleaning is scheduled, remove the bundle with the certified bundle puller and jet the outside of the bundle on the wash pad, taking care not to bend tube ends or displace baffle spacers.

6.6 Retain a sample of the removed deposit in a labelled container and estimate the volume removed. Deposit analysis distinguishes cooling water scale from process side polymer and directs the correct treatment response.

6.7 Inspect a minimum of ten percent of tubes by borescope after cleaning. Look for pitting, grooving at baffle locations, and inlet-end erosion. Report wall loss findings to the static equipment engineer.

6.8 Renew all gaskets with the material and rating specified on the drawing. Reused gaskets are not permitted under any circumstance.

6.9 Reassemble the heads and torque the flange studs in the star pattern to the values in the bolting table, in a minimum of three passes at thirty, seventy and one hundred percent.

6.10 Before the leak test, verify every temporary blank and spade installed for the work has been removed and is accounted for on the isolation certificate. A forgotten spade discovered at recommissioning costs the full outage again.

6.11 Perform a leak test at normal operating pressure during recommissioning, checking every disturbed joint with the shell and channel vents cracked to displace air.

6.12 Return the exchanger to service slowly, warming through over at least thirty minutes to avoid thermal shock to the tubesheet joints. During warm through, walk the unit and check each disturbed joint; a joint that seeps during warm up will usually reseal as the gasket beds, but any joint that streams shall be reported and the unit taken off load before retorquing. Record the post-cleaning terminal temperatures at steady state.

7. Acceptance Criteria

Cleaning is accepted when all scheduled tubes pass the lance freely, the borescope sample shows no defect requiring engineering disposition, the reassembled unit is leak free at operating pressure, and the approach temperature has recovered to within ten percent of the clean design value.

8. Records

Complete the inspection record with the as-found fouling estimate, tubes plugged, gasket batch numbers and the post-return thermal performance readings. File the record against the exchanger tag with a valid-until date per the inspection schedule. Retain the deposit sample analysis with the record; successive analyses justify changes to the cleaning interval and to the chemical treatment

programme.